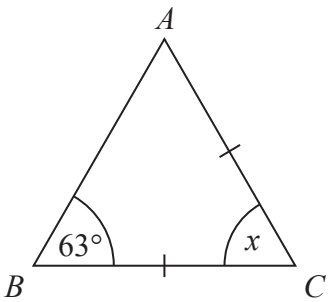
15 Mary needs to work out the size of angle x in this diagram.



She writes

$x = 63°$ because base angles of an isosceles triangle are equal.

Mary is wrong.

1 mark
x is not a base angle

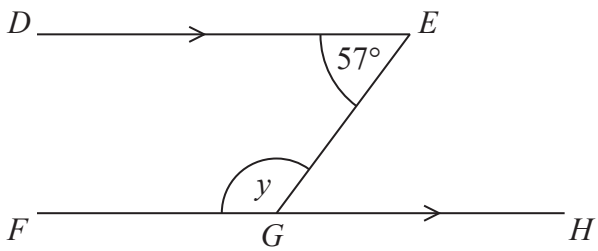
(a) Explain why.

AC = BC, so the equal base angles are A and B.

Angle x is between the equal sides; x = 180° – 126° = 54°.

(1)

William needs to work out the size of angle y in this diagram.



William writes

Working	**Reason**
angle $EGH = 57°$	because corresponding angles are equal
$y = 180° - 57°$ $y = 123°$	because angles on a straight line add up to 180°

One of William's reasons is wrong.

(b) Write down the correct reason.

Alternate angles between parallel lines are equal.

1 mark
This gives angle EGH = 57°

(1)

(Total for Question 15 is 2 marks)